Matheus Hoffmann

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👋 me

sr. software engineer & jr. researcher with **8 years** of experience building scalable (and occasionally clever) solutions mainly for payment processing & fraud-fighting, now building open-source ocean tech at **marola-dev**

🌊 open source — marola-dev

i'm actively building **marola** in the open: a citizen-science **ocean intelligence layer** for the brazilian coast. it answers *"can i swim tomorrow, and when?"* for every beach around florianópolis, rio de janeiro and salvador: the best hour from swell, wind, tide, uv and the official bathing-water bulletins, with the *why* in plain language, live at **marola.dev**.

- **water quality first** - every official sampling point (ima/sc, inea, inema) is on the map, coloured by its last bulletin; unfit water zeroes the score in deterministic scala 3 (kyo) code, and the llm explains but never overrules
- **the sea in front of you** - tide, swell, wind, uv, jellyfish and whale odds per beach, a sourced *did you know?* about the local sea, and *ask the ocean*: rag with citations on ollama, an mcp tool server and its own small model, marola-sea, no account or api key
- **designed in public** - every non-trivial change starts as a public design doc (MIPs); next up: wind, wave, satellite and sea-heat map layers, *ressaca* (storm surf) warnings, wave-model ensembles with WAVEWATCH III and an open data lake of beaches and water quality

🌊 marola.dev | FOSS marola-dev | docs docs.marola.dev | license MIT | DOI 10.5281/zenodo.23224155

civic-tech | disaster-risk-management | forecasting | meteorology | non-profit | oceanography | water-quality

> 🔓 contributors welcome, humans and ai agents alike — no code needed: use the map and tell us where it's wrong, share local sea knowledge, or open an issue or a PR at **marola-dev/marola**. or mail me: **mhoffmannfs[at]gmail.com**

⚡ research — ww3-gpu

my undergraduate project at ufrj (escola politécnica), co-advised at labeco/ufsc: **ww3 gpu lab**, an open lab for running noaa's **WAVEWATCH III** wave model and moving its expensive kernels to gpus *without changing the answer*. a nix-pinned fortran/mpi/netcdf toolchain builds ww3 and runs a regression test in one command. the dia nonlinear-interaction kernel (`W3SNL1`), ported to c++/**kokkos**, matches the fortran bit for bit on serial, openmp and cuda, and runs 1,000 sea points in 0.047 ms on an rtx 4090 against 24.96 ms serial. it also holds a 16-lesson course, from the wave spectrum to porting kernels with coding agents.

research ww3-gpu | DOI 10.5281/zenodo.23221351

💼 exp-highlights

- **itv (uk)** - ml/mlops on aws for channel audience forecasting (2025/aug 2026)
- **signifyd (us - award winner tenacious feb/2024)** - ml eng. for loss forecasting & risk-analysis on databricks (2023/2024)
- **elemeno AI (us - br client bvmf: AMER3)** - mlops engineer with k8s on gcp (2021/2022)
- **broad (uk - YC W21)** - sr. software engineer with scala for digital banking on gcp (2020)
- **stone (br: nasdaq: STNE)** - ml & software engineer with scala on aws (2017-2019)
- **LIOC (br: UFRJ/COPPE)** - ocean instrumentation lab - scientific research, ROV, microcontrollers (2016/2017)

🏅 certs

AWS Cloud Practitioner | AWS ML Specialty | AWS ML Engineer Associate | AWS Data Engineer Associate

📚 learning

GCP Professional Agentic Architect (studying)

🤿 misc

Strava open water swimming

PADI Advanced Open Water Diver | PADI Enriched Air Nitrox Diver | PADI Rescue Diver